PURDUE GIS DAY 2026

References and data sources

Mapping Vertiport Suitability in Austin, Texas

How Access Data and Planning Weights Shape the Shortlist

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This bibliography supports the research poster and its underlying Austin vertiport-siting study. It includes the supporting literature, planning guidance and source inventories, including the sources for all 14 EONS indicators and the study-area map. Data vintages and recorded access dates refer to the study inputs, rather than the latest available releases.

47 literature, methods and guidance references; 28 spatial-data and regional-context references. DOI and web links are clickable.

Source guide

Code	Input / purpose	Source inventory
E1	Population and primary-job density	Census ACS B01003; LODES 8 WAC, 2023, S000/JT01; 2024 TIGER/Line block groups
E2	Park-and-ride access	Capital Metro (2025); City of Austin (2016) parking only in the combined-source sensitivity run
O1	Aviation-facility proximity	FAA NASR, effective July 9, 2026
O2	Airspace class	FAA Aeronautical Information Services (2026)
O3	Emergency response	City of Austin fire stations (2026f)
O4	Road access	TxDOT roadway inventory (2026)
N1	Protected areas	USGS GAP: PAD-US 4.1
N2	Flood hazard	FEMA National Flood Hazard Layer (2026)
N3	Aquifer recharge	TCEQ Edwards Aquifer recharge zone (2005)
N4	Soil suitability	Soil Survey Staff: Texas gSSURGO, FY2026 release
S1	Transport-disadvantaged exposure	Census ACS B25044, 2020-2024 estimates
S2	Sensitive receptors	Texas Education Agency (2025); Texas DSHS (2025)
S3	Cultural sites	City of Austin historical landmarks and historic districts (2026g, 2026i)
S4	Public facilities	City of Austin centres and libraries (2026a-2026d); Capital Metro (2025)
Map	Study area and regional context	Austin jurisdiction (2026e), lakes/rivers (2026h), TxDOT, FAA NASR and CAMPO (2025)

Source keys use the author-year labels in the study bibliography. The 5 NM airport-core circle, 500 m infrastructure buffers and 1.0 km² shortlist criterion are model-defined planning assumptions, not universal vertiport requirements.

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